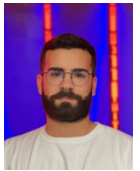


GUILLERMO JAVIER SIERRA-MAÍLLO MARTÍNEZ

INDUSTRIAL AND AUTOMATIC ELECTRONICS **ENGINEER** SPECIALIZED IN ROBOTICS



PERSONAL INFORMATION

Date of birth: 03 - 04 - 1995
Nationality: Spanish
Mobile phone: +34 622646261
Email: gsierra95@gmail.com
LinkedIn: @guillesierra

PROFILE

Experienced in real-time control, distributed systems and industrial automation across energy, railway and robotic sectors. I enjoy designing and delivering robust, practical solutions that bridge software and hardware, working both independently and in multidisciplinary teams.

PROFESSIONAL EXPERIENCE



Business Technology Consultant

MINSAIT, **JAN 2024 - PRESENT**

- Real-time Control & Operation Engineer in the **Energy Distribution** sector, delivering critical projects on the Onesait Intelligence Platform.
- Design and development of advanced **HMI**s for configuration, visualization and real-time monitoring of power grid assets.
- Architecture and integration of **distributed services** ensuring high availability and interoperability across heterogeneous systems and protocols.
- Collaboration with **cross-functional teams** to deploy solutions on **hybrid-cloud** environments following modern **DevOps** and reliability practices.



Systems Engineer

CAPGEMINI, **APR 2023 - JUN 2023**

- Developed and validated critical embedded software for **Railway Management Systems** with CAF Signalling.
- Implemented backend components in C/C++ with automated unit testing using gTest.
- Ensured compliance with railway sector safety and reliability standards throughout the development lifecycle.



ATM Systems Engineer

INDRA SISTEMAS, **JAN 2022 - APR 2023**

- Development of the integrated Network Management (INM) platform for **Air Traffic Management** systems.
- Participated in the **next generation** of network management operational systems for EUROCONTROL.
- Helped improve **safety** and **efficiency** of air transport in Europe through reliable monitoring and control solutions.



Connectivity and Cyber-Security Engineer

ASTI MOBILE ROBOTICS, **MAY 2021 - JAN 2022**

- Designed and implemented embedded 5G communication solutions for **Automatic Guided Vehicles** and **Autonomous Mobile Robots**, including smart charging stations.
- Developed **connectivity** architectures between vehicles, edge devices and cloud services, focusing on reliability and low latency.
- Performed **cybersecurity** risk analysis and implemented secure-by-design measures to protect networks, cloud servers, mobile devices and proprietary software.
- Contributed to guidelines and best practices for secure industrial communication and data protection.



R&D Development Engineer

ASTI MOBILE ROBOTICS, **AUG 2019 - JAN 2022**

- Programmed **automatic guided vehicles** (AGVs) and Yaskawa industrial and collaborative **robot arms** using proprietary software and ROS.
- Developed **embedded applications** in C/C++ and Python to control PLCs, implementing logic in Codesys V2.3 and V3.5.
- Designed and managed industrial communications using TCP/IP, UDP, OPC UA, CANBUS and CANopen.
- Built **industrial simulations** and digital twins of AGVs and robot arms with Visual Components.
- Participated in European **R&D projects**, handling development, commissioning, technical documentation and presentations with public and private partners.
- Maintained direct contact and negotiation with **clients** and **suppliers** to align technical solutions with business requirements.

SKILLS

Languages

- Spanish:** Native.
- English:** Advanced.
- Italian:** Basic.

Software

- Programming & scripting:** C, C++, Python, Go, JavaScript, Pug, HTML5, CSS3.
- Frameworks & platforms:** Vue 2.7, gRPC, REST APIs, ROS.
- Messaging & data:** NATS, Redis, PostgreSQL, MinIO.
- Cloud & DevOps:** AWS, Docker, hybrid-cloud architectures, unit testing with gTest, Linux, Bash, Git, CI/CD basics.
- Industrial & robotics tools:** Codesys V2.3/V3.5, RobotStudio ABB, Visual Components, MotoPlus, MotoSim, Yaskawa environments.
- Engineering software:** Altium Designer, Arquimedes, AutoCAD, Matlab, MPLAB X, PSIM, Quartus II, MES Visual Studio, Visual Studio Code, SparkAR.
- Office tools:** MS Word, MS PowerPoint, MS Excel, MS Access, MS Project, MS Teams.

UNIVERSITY EDUCATION

Master in Project Management

UNIVERSIDAD DE OVIEDO, **2022 - 2024**

- Interuniversity Master's course:** Given by the Universities of Oviedo, la Rioja and the Public University of Navarra.
- Credits:** 60 ECTS.

Bachelor of Industrial Electronics and Automatic Engineering (Bilingual)

UNIVERSIDAD DE OVIEDO, **2015 - 2019**

- Activities and associations:** Member of AEGEE European students forum. Mentor in the a-2UO program for new Erasmus students.
- TFG:** Analysis and processing of digital images for biometric purposes.
- Credits:** 240 ECTS.

UNIVERSITÀ DI BOLOGNA, **2016 - 2017**

- Activities and associations:** Member of Erasmus Student Network (ESN)
- Erasmus year.**

FURTHER TRAINING

Indra courses

INDRA SISTEMAS, **2022**

- ADA Programming Course** - Ivan Osuna Ayuste 44h.

Udemy courses

UDEMY, **2022**

- DevOps, CI/CD (Continuous Integration/Delivery)**
Cred. ID: UC-28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Docker for the Absolute Beginner - Hands On - DevOps**
Cred. ID: UC-2872c5c1-4773-4553-af90-agg430ee9ac5886
- Learning to work with Emacs**
Cred. ID: UC-28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Linux Shell Scripting: A Project-Based Approach**
Cred. ID: UC-28baf3daa-c07e-4e03-8f11-8eaa352b13d

Google courses

GOOGLE, **2021**

- Cloud computing.**
Cred. ID: 28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Digital skills for professionals.**
Cred. ID: 28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Fundamentals of Digital Marketing.**
Cred. ID: 28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Personal Productivity.**
Cred. ID: 28baf3daa-c07e-4e03-8f11-8eaa352b13d
- Digital transformation.**
Cred. ID: 28baf3daa-c07e-4e03-8f11-8eaa352b13d

Pluralsight courses

PLURALSIGHT, **2020**

- Build, Train, and Deploy Neural Networks with TensorFlow**
- Understanding Machine Learning with Python**

Language courses

CAMPUS SCHOOL OF ENGLISH, **2008 - 2013**

- English course** by Caroline Cogle

OTHER INFORMATION

- Driving license** (Spain and UEE): Category B. Own vehicle.
- Occupational Risk Prevention Course: Quirónprevención
- Occupational Risk Prevention Course: ANANDA

INTERESTS

Passionate about **technology**, **automotive** engineering and **design**. I regularly develop personal projects where I test innovative technologies and apply them to everyday solutions, while also exploring personal finance and investing.